Assignment 1

Problem 1: Reading the Shape of a Sample

Results

Using the Week 1 conventions, the sample has 1,000 observations:

- Mean: 0.0011043943
- Variance: 0.0000962482 (using n - 1)
- Skewness: -0.668831
- Excess kurtosis: 2.339849

(a) Predict

The negative skewness and positive excess kurtosis make NIG plausible. Normal is ruled out by both moments. Lognormal is ruled out because its skewness should be positive. Student's t is ruled out because it is symmetric.

(b) Fit

The fitted Normal has mean 0.0011043943, variance 0.0000962482, and standard deviation 0.0098106178. Its 1% quantile is -0.0217185155.

- Observed below the 1% quantile: 26
- Expected under the Normal: 10

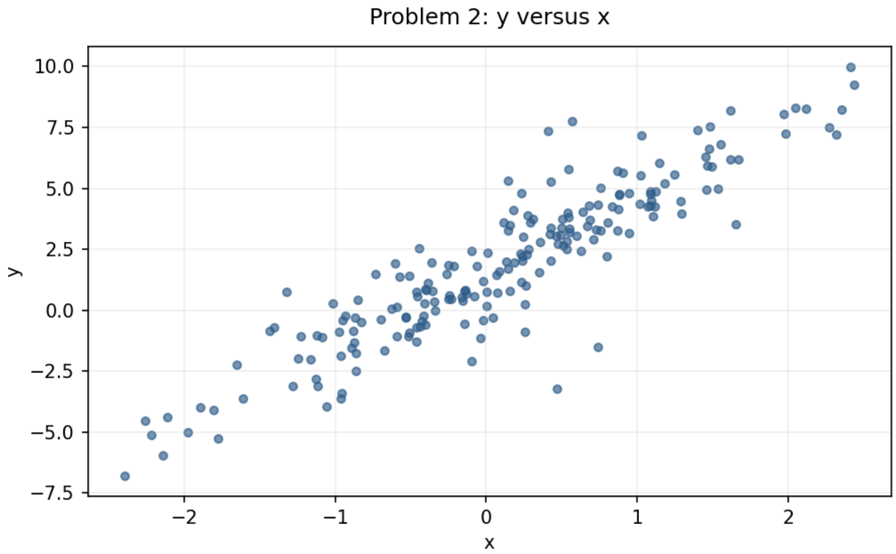
(c) Reconcile

The Normal underestimates left-tail risk. It predicts 1% below the cutoff, but the sample has 2.6%. The sample is more left-skewed and heavy-tailed than the fitted Normal.

Problem 2: A Regression Whose Errors Are Not Normal

(a)-(b) Predict

The scatter is strongly linear but has several large vertical outliers, so I expect Student's t errors. This violates OLS assumption 7: Normal errors. The slope remains unbiased and consistent if assumptions 1-6 hold, but heavy tails make it less efficient and outlier-sensitive. Its estimated standard error should be larger and less stable, and exact Normal-based inference fails.



Problem 2 scatter plotted before model fitting.

Fit

- OLS: alpha 1.560740 (SE 0.096269), beta 2.990757 (SE 0.096640), residual SD 1.351401
- Normal MLE: alpha 1.560740, beta 2.990757, scale 1.344627, AICc 692.145
- Student's t MLE: alpha 1.547980, beta 3.019179, scale 0.935509, df 3.632160, AICc 665.327

(c)-(e) Reconcile

The slopes are close, so non-Normal errors did not move the slope much. Student's t wins by 26.817 AICc points because the Normal misses the heavy tails.

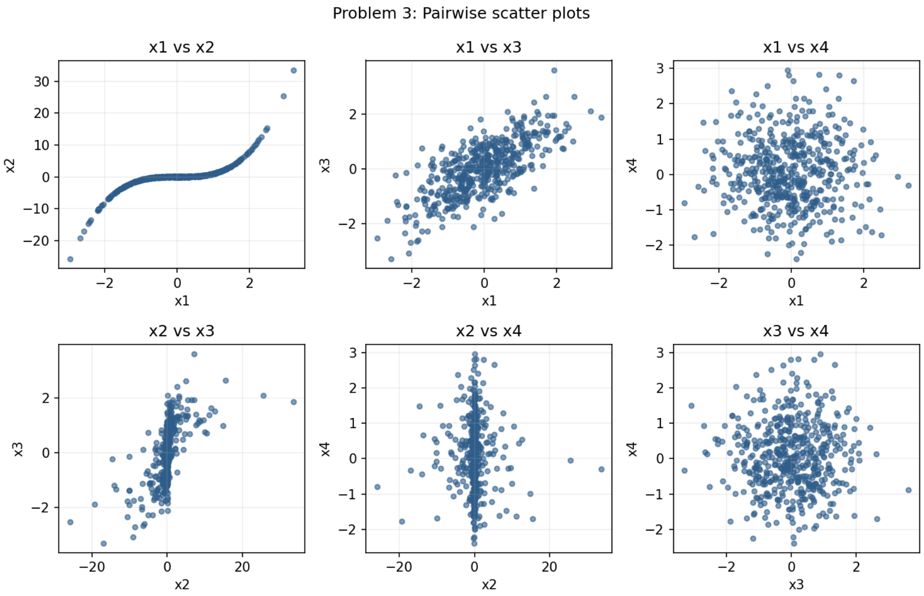
- 95% error quantile: Normal 2.211715; Student's t 2.053833
- 99.5% error quantile: Normal 3.463530; Student's t 4.620089

The Normal is wider at 95% because its fitted scale is larger, but the t is wider in the far tail. I would use the t model for a capital buffer.

Problem 3: Pearson Against Spearman

(a) Predict

I expect the largest disagreement for x1 and x2 because their relationship is strongly monotonic but curved. The measures should be closer for the roughly linear x1-x3 pair and for pairs involving independent x4.



Pairwise scatter plots for all six variable pairs.

(b) Fit

Pearson correlation matrix:

	x1	x2	x3	x4
x1	1.000000	0.799092	0.719467	-0.004680
x2	0.799092	1.000000	0.587614	-0.020923
x3	0.719467	0.587614	1.000000	-0.008605
x4	-0.004680	-0.020923	-0.008605	1.000000

Spearman correlation matrix:

	x1	x2	x3	x4
x1	1.000000	0.971930	0.683910	0.000189
x2	0.971930	1.000000	0.667422	-0.001131
x3	0.683910	0.667422	1.000000	-0.002422
x4	0.000189	-0.001131	-0.002422	1.000000

The largest gap is x1-x2: Pearson 0.799092, Spearman 0.971930, gap 0.172838.

(c) Reconcile

The curve keeps almost the same ranking but is not a straight line. Spearman better describes its monotonic strength; Pearson describes linear strength.

Problem 4: Conditional Distributions

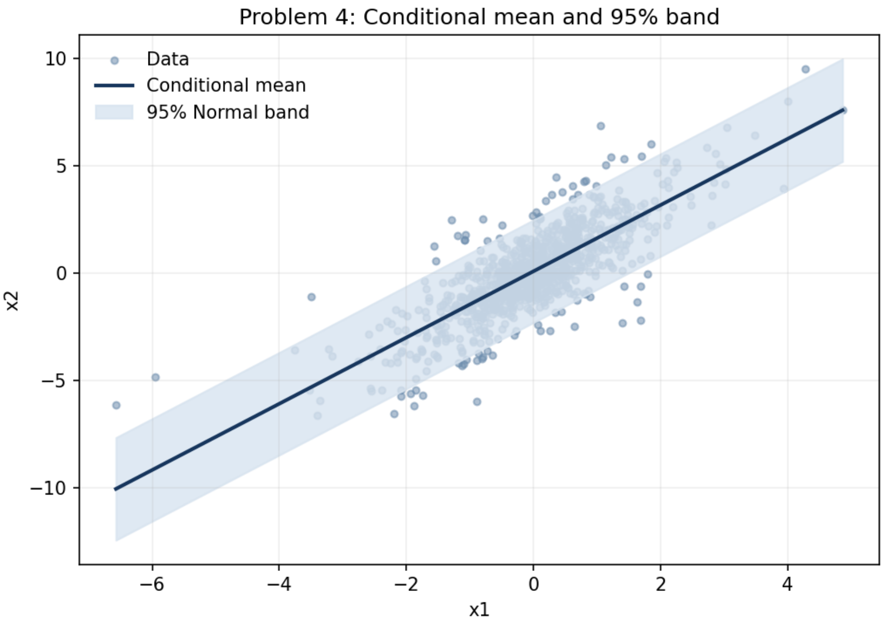
Following Week 2 notation, block X1 is target x2 and block X2 is observed x1. Thus Sigma11 = Var(x2) = 4.104749, Sigma22 = Var(x1) = 1.099875, and Sigma12 = Sigma21 = Cov(x2,x1) = 1.696902.

(a)-(b) Predict

Var(x2 | x1) = Sigma11 - Sigma12 × Sigma22^-1 × Sigma21 = 1.486746. The remaining variance factor is (Sigma11 - Sigma12 × Sigma22^-1 × Sigma21) / Sigma11 = 0.362201, a 63.78% reduction. The remaining standard-deviation factor is 0.601832. Under the multivariate Normal, this does not depend on observed x1.

(c) Fit

E[x2 | x1=a] = mu1 + Sigma12 × Sigma22^-1 × (a - mu2), where mu1 is mean(x2) and mu2 is mean(x1). The coefficient on x1 is 1.542813, the OLS slope. The line is 0.071073 + 1.542813a. The constant 95% band has half-width 2.389827.



Conditional mean and the constant-width 95% Normal band.

(d)-(f) Reconcile

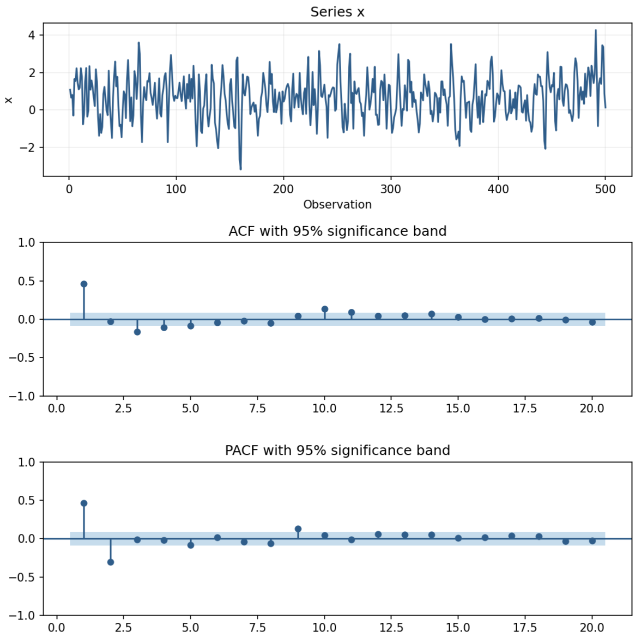
- Overall coverage: 93.50%
- Within 1 SD of mean x1: 95.65% (759 observations)
- Between 1 and 2 SD: 85.79% (190 observations)
- Beyond 2 SD: 90.20% (51 observations)

Coverage is not flat, so conditional variance is not constant. The OLS slope remains the best linear coefficient, but the exact Normal conditional distribution and constant-width band fail.

Problem 5: Identifying an AR or MA Order

(a)-(b) Predict

The ACF decays with an oscillating pattern. The PACF has large values at lags 1 and 2, then mostly cuts off. With an approximate 95% band of +/-0.087654, I predict AR(2): an AR(p) has a decaying ACF and a PACF that cuts off after p.



Series, ACF, and PACF for Problem 5.

(c)-(d) Fit and reconcile

- AR(1): AICc 1418.587
- AR(2): AICc 1372.438
- AR(3): AICc 1374.343
- MA(1): AICc 1389.084
- MA(2): AICc 1381.232
- MA(3): AICc 1374.117

AICc selects AR(2), matching the prediction.

(e) AR(2) versus AR(3)

AR(2) coefficients are 0.601201 and -0.302638. AR(3) coefficients are 0.596203, -0.292690, and -0.016542. The third coefficient is very small, so AICc rejects it after penalizing the extra parameter. R-squared does not penalize added variables and therefore would not make the same choice.